Introduction

The Weibull distribution generalises the exponential to allow monotone non-constant hazard rates. It is the default parametric model in reliability engineering and a common choice for parametric survival analysis. The shape parameter controls whether hazard is increasing (ageing), decreasing (early failures dominate), or constant (exponential).

Prerequisites

Exponential distribution, hazard and survival functions.

Theory

A Weibull random variable with shape $k > 0$ and scale $\lambda > 0$ has PDF

$$f(x) = \frac{k}{\lambda}\left(\frac{x}{\lambda}\right)^{k-1} \exp\left[-\left(\frac{x}{\lambda}\right)^{k}\right], \qquad x \geq 0.$$

CDF: $F(x) = 1 - \exp\big[-(x/\lambda)^k\big]$.

Hazard function: $h(x) = (k/\lambda)(x/\lambda)^{k-1}$. Three regimes:

- $k < 1$: decreasing hazard (early failures / "infant mortality").
- $k = 1$: constant hazard (exponential).
- $k > 1$: increasing hazard (wear-out / ageing).

Moments:

- $E[X] = \lambda\,\Gamma(1 + 1/k)$.
- $\mathrm{Var}(X) = \lambda^2\left[\Gamma(1 + 2/k) - \Gamma(1 + 1/k)^2\right]$.

Accelerated failure time form: $\log T = \mu + \sigma W$ where W is standard extreme-value; the AFT parameters relate to Weibull's $k = 1/\sigma$ and $\lambda = e^{\mu}$.

Assumptions

Non-negative continuous data, monotone hazard (log-hazard linear in log-time).

R Implementation

```r
library(survival)

# PDFs for three shape regimes
x <- seq(0, 5, length.out = 400)
plot(x, dweibull(x, shape = 0.5, scale = 1), type = "l", col = "#F4A261",
     lwd = 2, main = "Weibull PDFs", ylab = "f(x)")
lines(x, dweibull(x, shape = 1, scale = 1), col = "#6A4C93", lwd = 2)
lines(x, dweibull(x, shape = 2.5, scale = 1), col = "#2A9D8F", lwd = 2)
legend("topright", c("k = 0.5 (decreasing h)", "k = 1 (exponential)", "k = 2.5 (increasing h)"),
       col = c("#F4A261", "#6A4C93", "#2A9D8F"), lwd = 2)

# Simulate and fit
set.seed(2026)
X <- rweibull(500, shape = 2.0, scale = 1.5)
fit <- survreg(Surv(X, rep(1, length(X))) ~ 1, dist = "weibull")
summary(fit)
```

```r
# Convert survreg output to Weibull parameters
c(shape = 1 / fit$scale,
  scale = exp(coef(fit)))
```

Output & Results

```
Call: survreg(...)

                Value Std. Error     z       p
(Intercept)  0.40578    0.01574 25.78  <2e-16
Log(scale)  -0.69312    0.03293 -21.0  <2e-16

Scale = 0.500

   shape    scale
   2.000    1.501
```

The fit recovers $k \approx 2$ and $\lambda \approx 1.5$ – the true parameters. The increasing-hazard regime is visible in the PDF.

Interpretation

In survival analysis, a Weibull model is often the starting parametric fit: "we fitted a Weibull regression with shape $k = 1.3$ (95 % CI 1.1-1.5), indicating increasing hazard over follow-up." Weibull is the default choice when the proportional-hazards assumption holds on a transformed scale.

Practical Tips

- R's `rweibull(n, shape, scale)` conventions differ from `survreg`, which parameterises via intercept and scale $= 1/k$.
- For right-censored data, use `survreg(..., dist = "weibull")`, not `fitdistr` on complete data.
- Check the fit via a log-log plot: $\log(-\log S(t))$ vs $\log t$ should be approximately linear.
- If shape estimates are very close to 1 with narrow CI, consider the simpler exponential.
- `flexsurv::flexsurvreg(..., dist = "weibull")` is a modern alternative with richer diagnostics.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Probability, conditional probability, Bayes' theorem
- Discrete distributions: Bernoulli, binomial, Poisson
- Continuous distributions and Q-Q plots